Which Romanian LLMs Are the Best Feature Extractors?

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1. Introduction

Large language models are expensive to fine-tune fully. We instead use them as **frozen feature extractors**: extract hidden states, then train a lightweight MLP classifier on top. We ask two questions: (1) which Romanian LLM produces the most discriminative embeddings for sentiment classification, and (2) which pooling strategy best converts per-token hidden states into a single sentence embedding?

2. Dataset and Preprocessing

We use LaRoSeDA, a Romanian sentiment dataset of product reviews, labeled as **positive** or **negative**. The dataset has 15000 reviews, evenly split between the classes. We concatenate the title and review content, and then truncate it to 256 tokens.

3. Models

We evaluate four decoder-only LLMs as feature extractors: **LLaMA 3.1-8B**: a general-purpose English-dominant baseline; and **RoLLaMA3-8B-Instruct**, **mGPT-1.3B**, and **LLMic-3B** as Romanian models. 8B models are loaded in 4-bit NF4 quantization (bitsandbytes) to fit GPU memory. The backbone is frozen throughout; only the MLP classifier is trained.

4. Pooling Strategies

Standard LLMs produce one hidden state per token. We compare four strategies for reducing the sequence to a single sentence vector:

- **Last token:** hidden state of the final non-padding token
- **Mean pooling:** attention-mask-weighted average over all tokens
- **Echo:** input repeated twice; hidden state taken after second occurrence, allowing each token to attend to itself
- **Summary:** input wrapped as *"Acest text: {TEXT} inseamna intr-un cuvant:"*, last token taken as compressed semantic representation

All embeddings are L2-normalised before being passed to the MLP.

5. MLP Classifier

A shared MLP architecture is used across all experiments:

`input_dim` $\rightarrow$ `512` $\rightarrow$ `256` $\rightarrow$ `2`

Each hidden layer uses BatchNorm, GELU activation, and Dropout(0.3). Trained with AdamW (lr=1e-3, weight decay=1e-4) for 30 epochs. Best checkpoint selected by validation macro F1.

6. Results

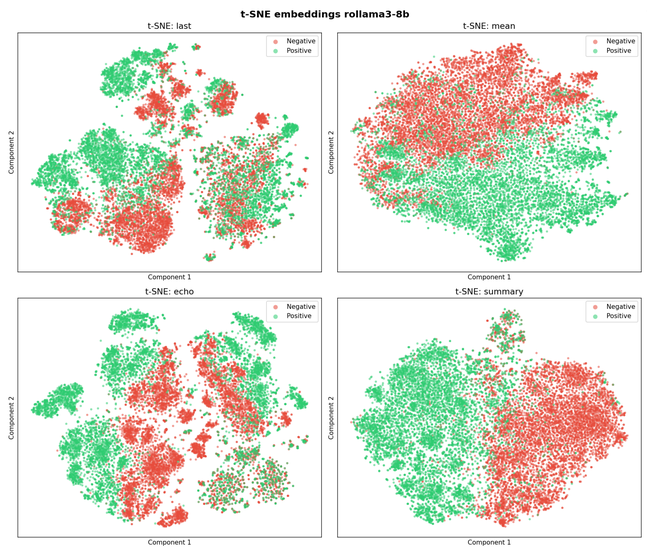
All four models' macro F1 scores are evaluated under all four pooling strategies.

Pooling strategy	Last Token	Mean Pooling	Echo	Summary
LLaMA 3.1-8B	0.946	0.974	0.937	0.955
RoLLaMA3-8B-Instruct	0.956	**0.975**	0.952	0.955
mGPT-1.3B	0.899	0.959	0.920	0.954
LLMic-3B	0.962	0.964	0.973	0.967

Mean pooling is consistently the strongest strategy across models, achieving the top two scores overall: RoLLaMA3-8B/mean (0.975) and LLaMA3.1-8B/mean (0.974). This is expected — averaging integrates sentiment signal distributed across all tokens rather than betting on a single position. **Echo** performs competitively for the 8B models but offers no consistent gain. **Last token** is weakest for mGPT (0.899), where the GPT-2 architecture may not aggregate context as effectively into the final token. mGPT is the most sensitive to pooling choice, with a 6-point gap between its best (echo, 0.959) and worst (last, 0.899) strategies.

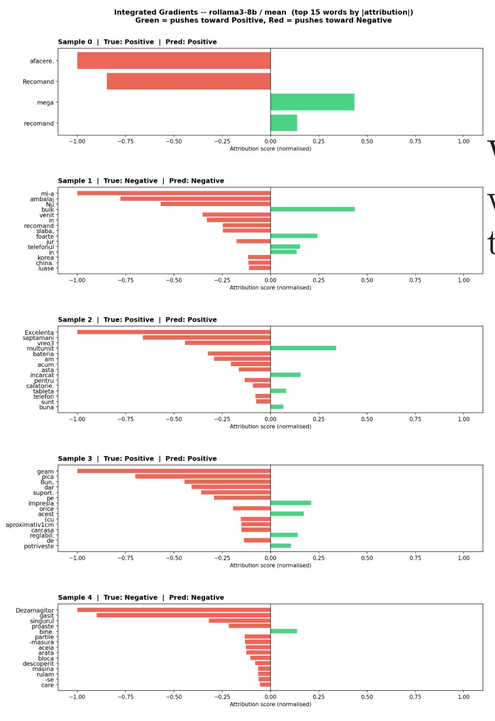
The Romanian models (LLMic, RoLLaMA), on average, match or exceed the general LLaMA baseline, confirming that language-specific pre-training adds value in a frozen feature extractor setting.

7. Analysis



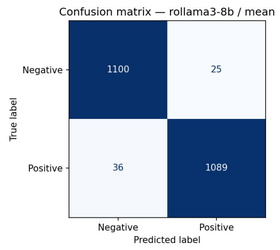
We plot the t-SNE embeddings of RoLLama (perplex.=30), color=sentiment.

- **Last** and **Echo** form distinct local clusters but with notable class mixing in the overlap regions.
- **Mean** produces a smoother, more uniformly distributed space with a cleaner class gradient, consistent with its superior downstream F1.
- **Summary** shows a coarser global separation but with class mixing within each region, suggesting the prompt template induces structure that does not fully align with sentiment.



We apply **Integrated Gradients** via the Captum library to identify which input tokens most influenced the classifier's predictions. **Green** tokens push toward Positive; **red** tokens push toward Negative.

- The model consistently assigns high attribution to *evaluative adjectives* (e.g. quality descriptors, ratings language) and low attribution to *structural tokens* (punctuation, conjunctions, articles).
- Misclassified samples tend to contain ambiguous sentiment language or mixed positive/negative phrases within a single review.



We compute the confusion matrix on the test set for the best configuration (RoLLaMA3-8B + mean pooling, macro F1 = 0.975). Both classes are classified with near-equal precision and recall, consistent with the balanced dataset and high macro F1.

8. Conclusions

- Romanian-specific LLMs match or exceed a general LLaMA 3.1-8B baseline as frozen feature extractors on Romanian sentiment classification.
- **Mean pooling** is the most robust strategy across all models and should be preferred over last-token pooling for sentence-level tasks.
- **Echo and Summary** embeddings offer no consistent improvement and add inference cost (longer sequences). Integrated Gradients confirm the model focuses on semantically meaningful tokens, lending interpretability to the pipeline.

References

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